

SIMATS ENGINEERING
Department of Computer Science and Engineering

COURSE NAME:	Cloud Computing and Big Data Analytics – Model Practical Examination	COURSE CODE:	CSA1522
---------------------	--	---------------------	---------

Name	KEERTHANA.I
Register Number	192521062

MODEL PRACTICAL EXAMINATION – SET 6

Answer all questions. Each question carries equal marks (20).

Total marks: 100

1. Create a simple cloud software application for Online Super Market using any Cloud Service Provider to demonstrate SaaS, with the template of Customer Name, Address, Phone, Email, Items, Quantity, etc. along with the item classification form.

AIM:

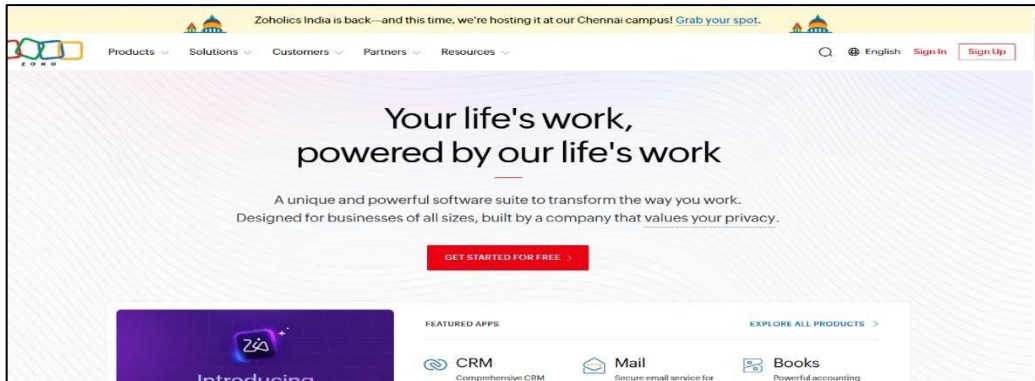
To create a simple cloud-based Online Super Market application using Zoho Creator to demonstrate the Software as a Service (SaaS) model.

PROCEDURE

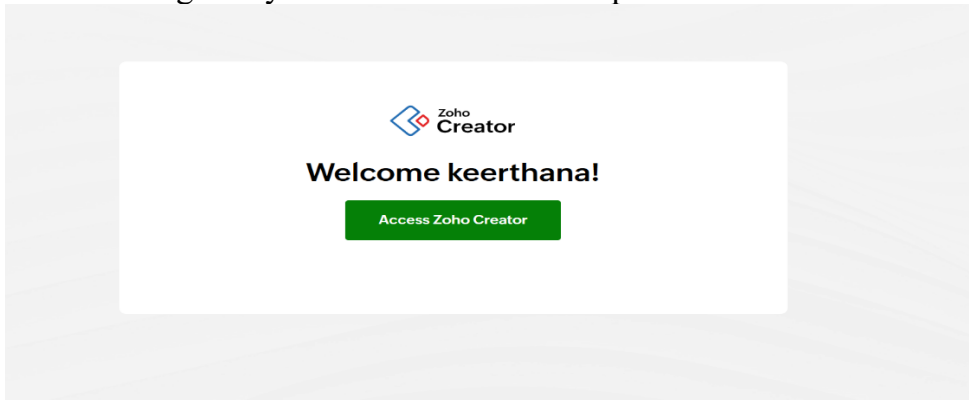
1. Login to Zoho Creator and create a new application.
2. Create **Customer Details, Item Classification, and Order Details** forms.
3. Add the required fields and enter sample records.
4. Create **Reports and Dashboard** for the application.
5. Test the application and verify the output.

Implemented

Step 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator



STEP 2: Create a New Application

Create a new application and enter **Online Super Market Management System** as the application name.

Create from scratch

Application Name *

Super market management system

☐ Enable Environment [Learn More](#)

Create Application

STEP 3: Create Customer Details Form

Create the Customer Details form and add the required customer fields.

192521212.simats_saveetha - S x +

creatorapp.zoho.in/192521212.simats_saveetha/super-market-management-system#Form:customer_details

Super Market Management System Trial expires in

Super Market Managemen...

Dashboard

Order Details >

Sales Reports >

Customer Details >

+ Customer Details

All Customer Details

Item Classificatio... >

Items >

Sales Orders >

Customers >

Customer Details

Customer Name

Address

Address Line 1

Phone Number

+91 81234 56789

Email

Linked Contact

-Select-

Submit

Reset

STEP 4: Create Item Classification Form

Create the Item Classification form with item name, category, price, and available quantity.

192521212.simats_saveetha - 5 X +

creatorapp.zoho.in/192521212.simats_saveetha/super-market-management-system#Form:items

Super Market Management System Trial expires in 15 days

Super Market Management System

- Dashboard
- Order Details
- Sales Reports
- Customer Details
- Item Classification
- Items**
 - + Items**
 - All Items
 - Sales Orders
 - Customers

Items

Brand	
Category	
Cost Price	#####.##
Created Time	dd-MMM-yyyy HH:mm:ss
Dimension Unit	
EAN	#####
Height	#####.##
HSN Or SAC Code	#####
ISBN	
Last Modified Time	dd-MMM-yyyy HH:mm:ss
Length	#####.##

STEP 5: Create Order Details Form

Create the Order Details form and add customer, item, quantity, price, total amount, order date, and payment status fields.

192521212.simats_saveetha - 5 X +

Ask Gemini

creatorapp.zoho.in/192521212.simats_saveetha/super-market-management-system#Form:order_details

Super Market Management System Trial expires in 15 days Upgrade Edit this application Help

Super Market Management System

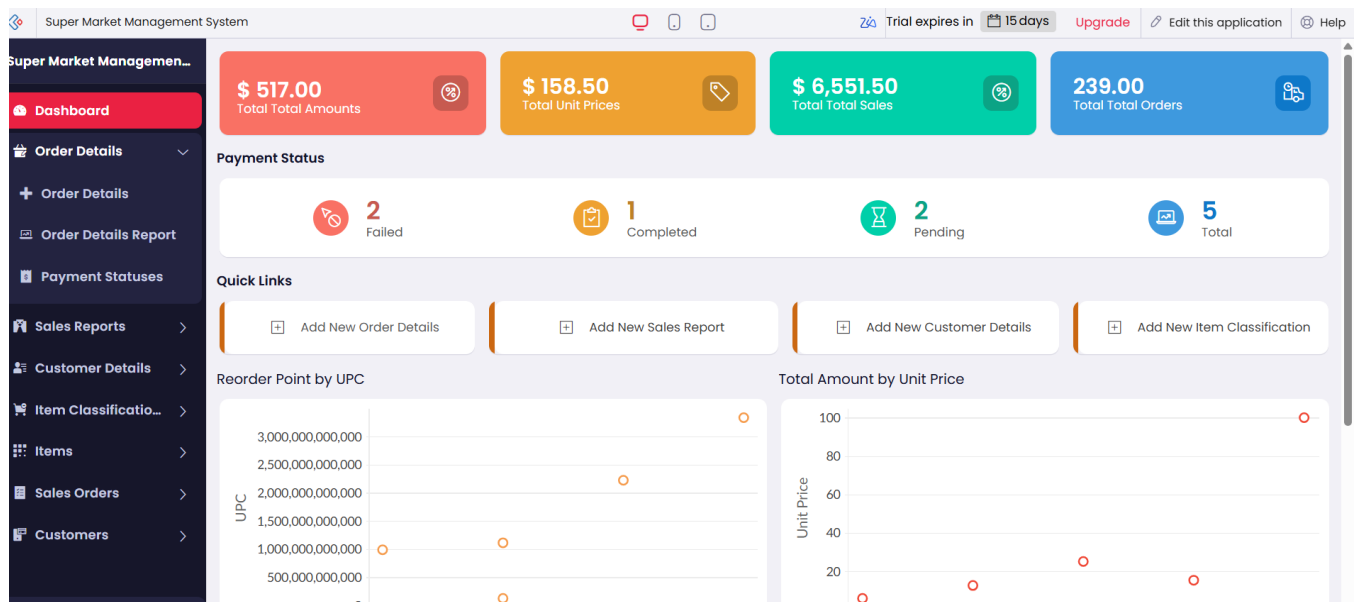
- Dashboard
- Order Details**
 - + Order Details**
 - Order Details Report
 - Payment Statuses
- Sales Reports
- Customer Details
- Item Classification
- Items
- Sales Orders
- Customers

Order Details

Order Date	dd-MMM-yyyy
Quantity	#####
Unit Price	#####.##
Total Amount	#####.##
Payment Status	-Select-
Customer	-Select-
Item	-Select-
Linked Sales Order	-Select-

STEP 6: Create Final Dashboard

Create the Online Super Market Dashboard with summary panels and charts.



RESULT

Thus, the **Online Super Market SaaS application** was successfully created using Zoho Creator with three forms, reports, dashboard, and cloud-based data storage.

2. Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS. Include the necessary fields such as Name of the Student, Register No., Subjects, Marks, Average, Percentage, Rank, Overall Performance, etc.

AIM

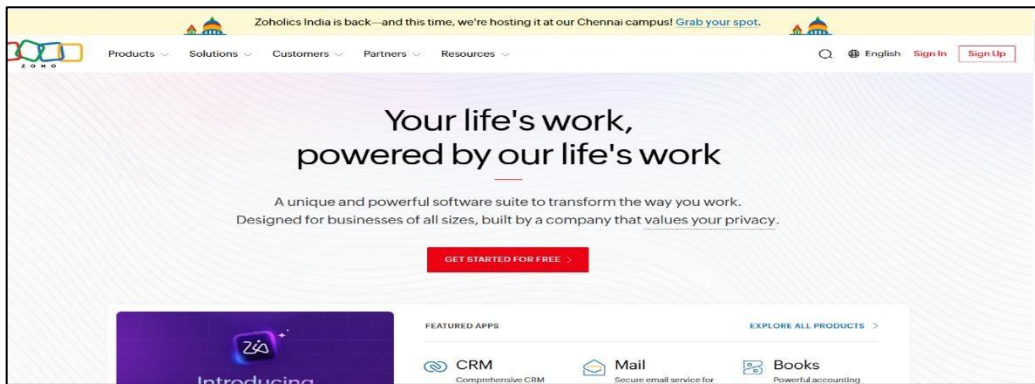
To create a cloud-based Mark Sheet Processing System using Zoho Creator to demonstrate the SaaS model.

PROCEDURE

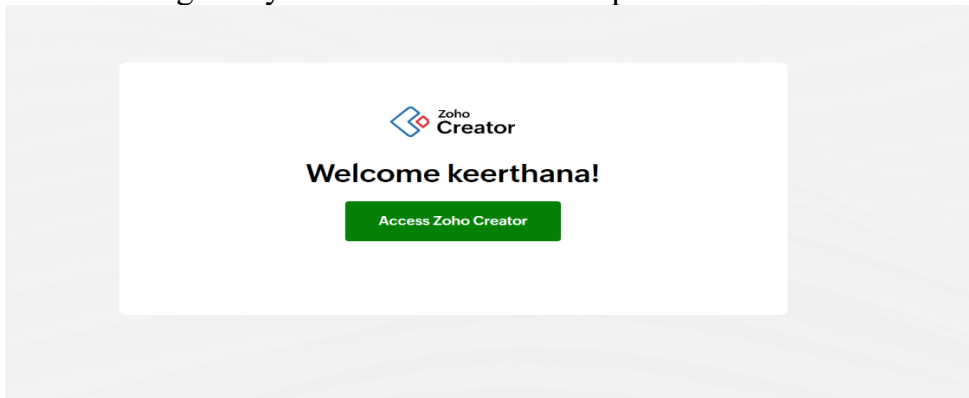
1. Login to Zoho Creator and create a new application.
2. Create Student Details, Subject Details, Mark Details, and Mark Sheet Summary forms.
3. Add the required fields and enter sample records.
4. Create Reports and Dashboard.
5. Test the application and verify the output.

STEPS

Step 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator



Step 3: Create a new application named **Mark Sheet Processing System**.

The screenshot shows the "Create from scratch" form in Zoho Creator. It has a header with the Zoho logo and the text "Create from scratch". Below this is a form with a label "Application Name *" and a text input field containing "college mark sheet processing". There is also a checkbox labeled "Enable Environment" with a link "Learn More". At the bottom is a blue button that says "Create Application".

Step 4: Create the **Student Details** form and add the required fields.

192521212.simats_saveetha - C x +

creatorapp.zoho.in/192521212.simats_saveetha/college-mark-sheet-processing-system#Form:student_details

College Mark Sheet Processing System

College Mark Sheet Proces...

- Dashboard
- Mark Details >
- Mark Sheet Summ... >
- Subject Details >
- Student Details v
- + Student Details**
- All Student Details
- Genders

Student Details

Student Name	
	First Name
Register Number	
Department	
Year	
Email Address	
Phone Number	+91 81234 56789
Address	
	Address Line 1
Date Of Birth	dd-MMM-yyyy
Gender	-Select-
Enrollment Status	-Select-

Step 5: Create the **Subject Details** form and add the required fields.

192521212.simats_saveetha - C x +

creatorapp.zoho.in/192521212.simats_saveetha/college-mark-sheet-processing-system#Form:subject_details

College Mark Sheet Processing System

College Mark Sheet Proces...

- Dashboard
- Mark Details >
- Mark Sheet Summ... >
- Subject Details >
 - + Subject Details
 - All Subject Details
 - Subject Types
- Student Details >

Subject Details

Subject Code	
Subject Name	
Maximum Marks	#####
Semester	
Subject Type	-Select-
Credits	#####
Instructor Name	
	First Name
Description	
Start Date	dd-MMM-yyyy

Step 6: Create the **Mark Details** form and add the required fields.

192521212.simats_saveetha - C x +

creatorapp.zoho.in/192521212.simats_saveetha/college-mark-sheet-processing-system#Form:mark_details

College Mark Sheet Processing System

College Mark Sheet Proces...

- Dashboard
- Mark Details >
 - + Mark Details
 - All Mark Details
 - Grades
- Mark Sheet Summ... >
- Subject Details >
- Student Details >

Mark Details

Student	-Select-
Subject	-Select-
Marks Obtained	#####
Maximum Marks	#####
Grade	-Select-
Result	-Select-
Exam Date	dd-MMM-yyyy
Exam Center	
Internal Marks	#####
External Marks	#####
Remarks	

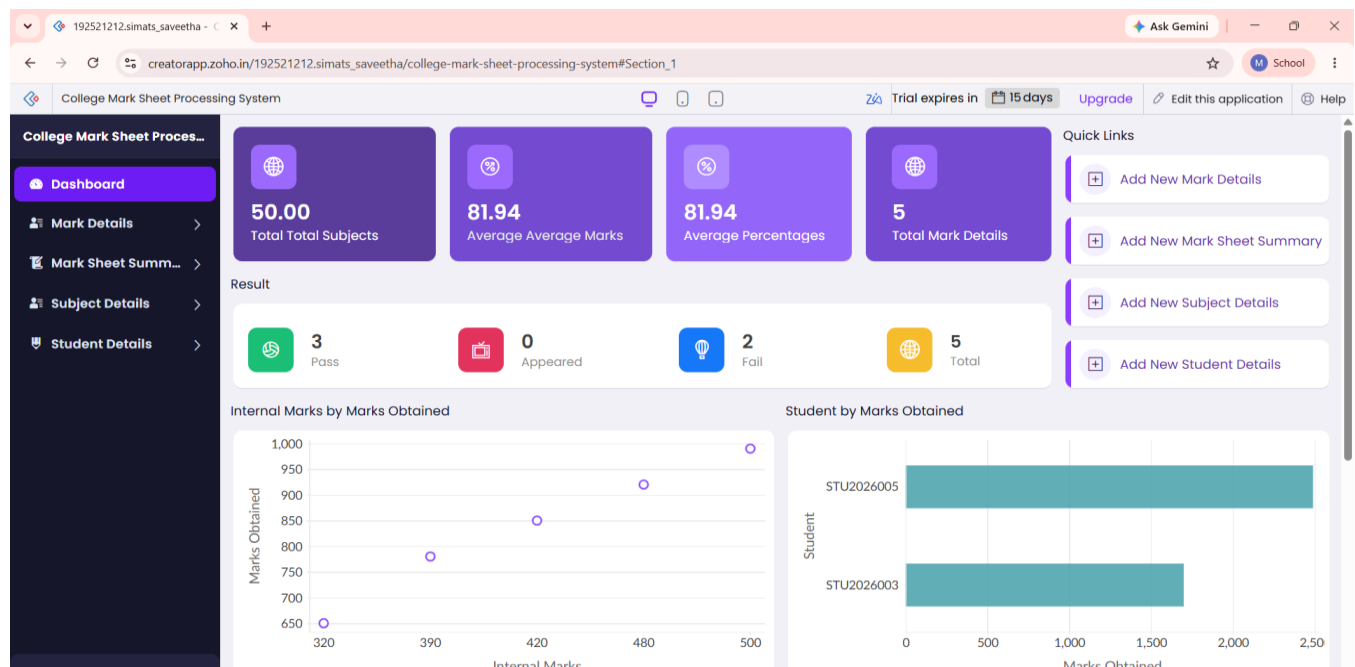
Step 7: Create the Mark Sheet Summary form and add the required fields.

College Mark Sheet Processing System

Mark Sheet Summary

Student	-Select-
Average Marks	#####.##
Percentage	
Rank	#####
Overall Performance	-Select-
Total Subjects	#####
Pass Count	#####
Fail Count	#####
Generated Date	dd-MMM-yyyy HH:mm:ss
Verified By	
Notes	

Step 8: Create the Dashboard.



RESULT

Thus, the Mark Sheet Processing System was successfully created using Zoho Creator with four forms, reports, and dashboard.

3. Install VirtualBox with different flavours of Linux or Windows OS on top of Windows 10 using custom installation.

AIM

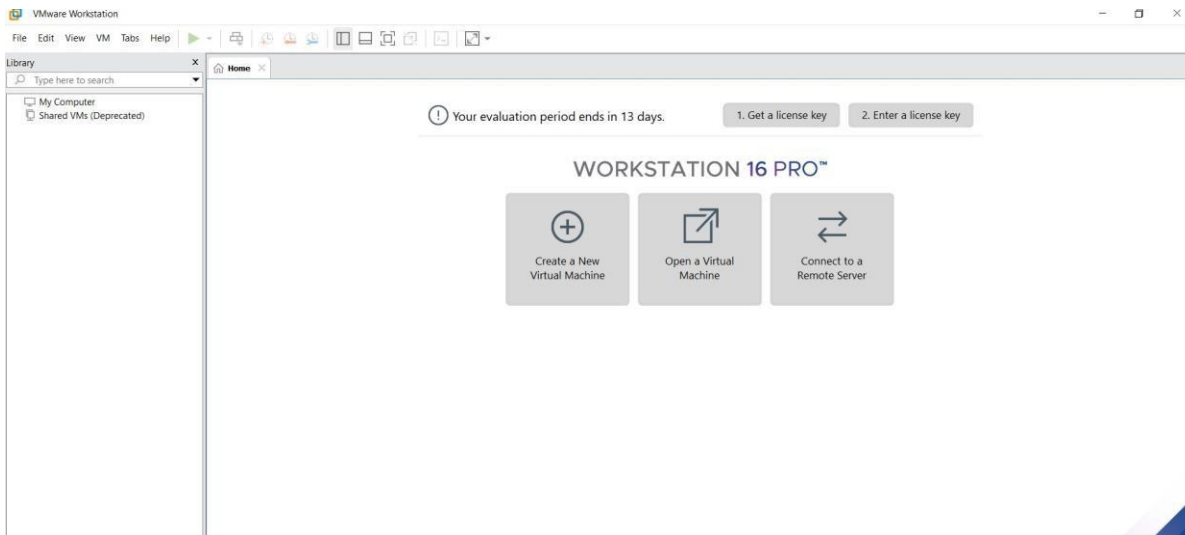
To demonstrate virtualization by installing a Type-2 Hypervisor and creating and configuring a virtual machine with a host operating system.

PROCEDURE

1. Download and install VMware Workstation as a Type-2 Hypervisor.
2. Download Ubuntu or Tiny OS as an ISO image file.
3. Open VMware Workstation and create a new virtual machine.
4. Perform the basic configuration settings.
5. Create the virtual machine and launch it.

STEPS

Step 1: Download and install VMware Workstation as a Type-2 Hypervisor.

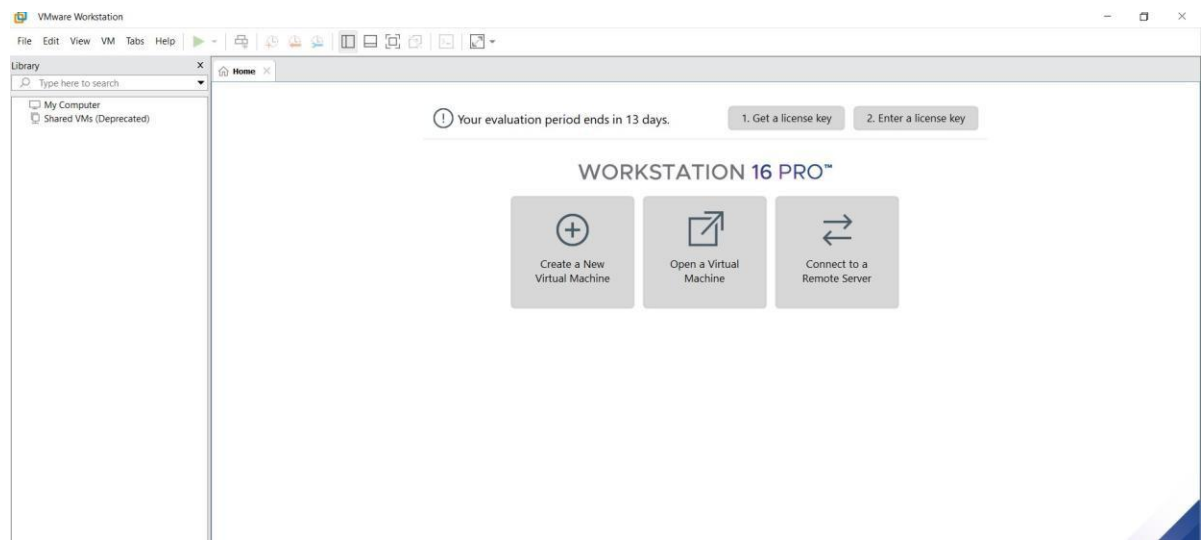


Step 2: Download Ubuntu or Tiny OS as an ISO image file.

Index of /11.x/x86/release/

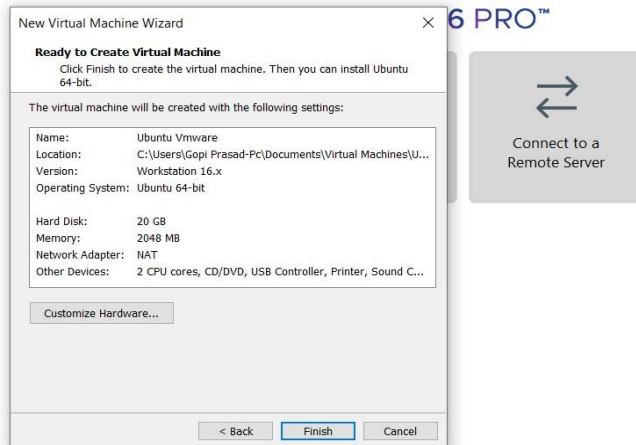
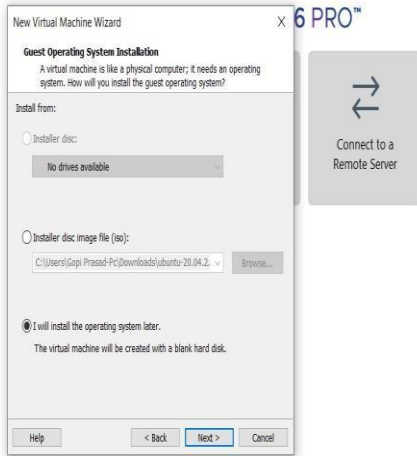
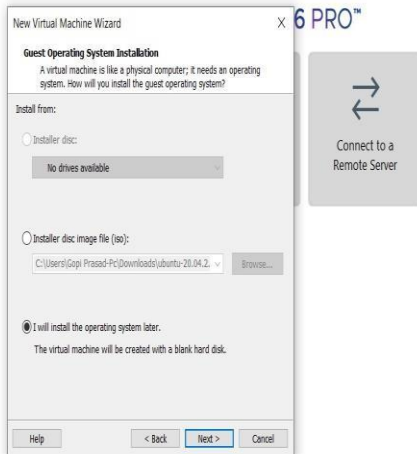
../	09-Feb-2020 11:50	-
distribution_files/	03-Dec-2019 11:14	-
src/	01-Apr-2020 07:49	14757888
Core-11.1.iso	01-Apr-2020 07:49	48
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	50639
Core-11.1.iso.zsync	01-Apr-2020 07:49	14757888
Core-current.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso	01-Apr-2020 07:50	52
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	369358
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	216006656
CorePlus-current.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso	01-Apr-2020 07:50	52
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	68301
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	19922944
TinyCore-current.iso		

Step 3: Open VMware Workstation and select Create New VM.

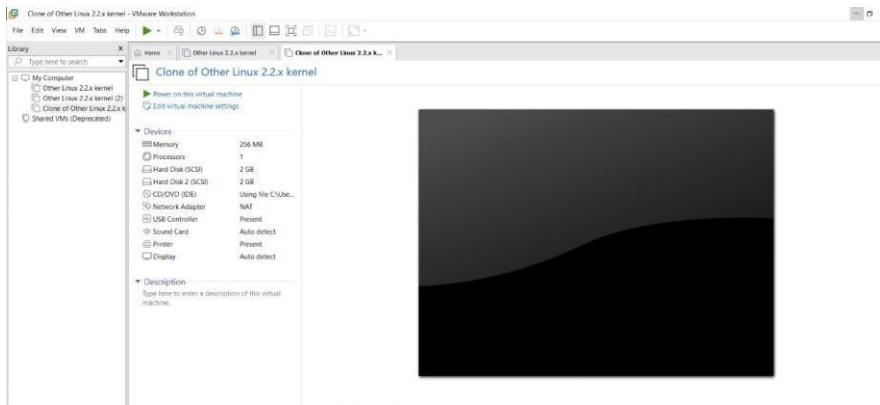


Step 4: Perform the basic configuration settings.

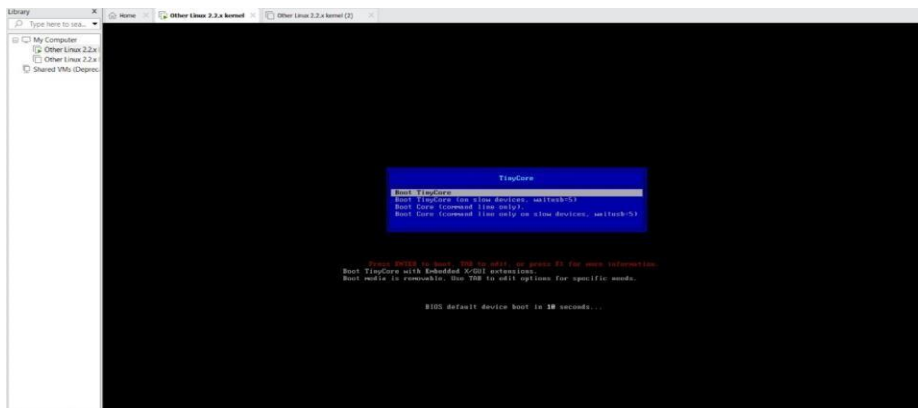




Step 5: Create the virtual machine.



Step 6: Launch the virtual machine.



RESULT

Thus, the virtual machine was successfully created and configured using VMware Workstation as a Type-2 Hypervisor.

4. Write a procedure to run virtual machines of different configurations and check how many virtual machines can be utilized at a particular time.

AIM

To run virtual machines with different configurations and check how many virtual machines can be utilized at a particular time.

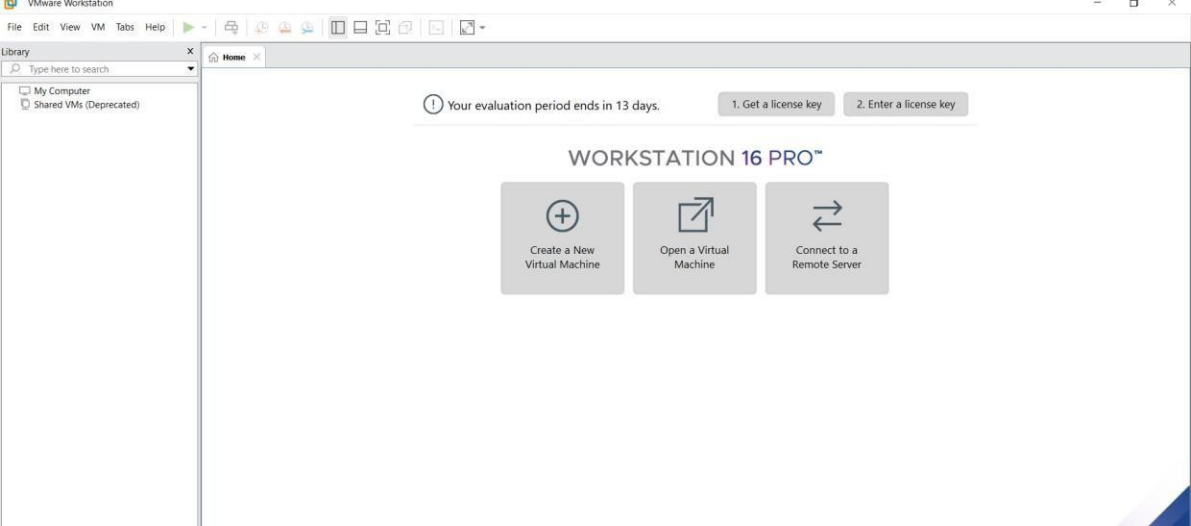
PROCEDURE

1. Open VMware Workstation and create virtual machines with different configurations.

- 2. Allocate different CPU, RAM, and storage to the virtual machines.
- 3. Start the virtual machines and observe their performance.
- 4. Run multiple virtual machines simultaneously.
- 5. Check the number of virtual machines that can run at the same time.

STEPS

Step 1: Open VMware Workstation and select the required virtual machine.

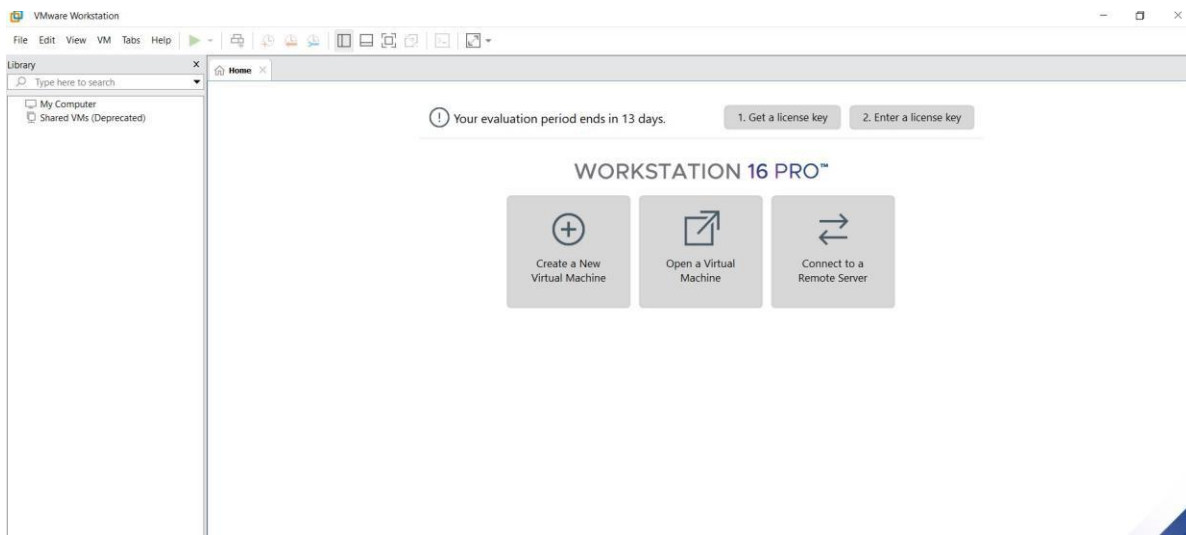


Step 2: Configure the CPU, RAM, and storage for the virtual machine.

Index of /11.x/x86/release/

../	09-Feb-2020 11:50	-
distribution_files/	03-Dec-2019 11:14	-
src/	01-Apr-2020 07:49	14757888
Core-11.1.iso	01-Apr-2020 07:49	48
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	50639
Core-11.1.iso.zsync	01-Apr-2020 07:49	14757888
Core-current.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso	01-Apr-2020 07:50	52
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	369358
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	216006656
CorePlus-current.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso	01-Apr-2020 07:50	52
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	68301
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	19922944
TinyCore-current.iso		

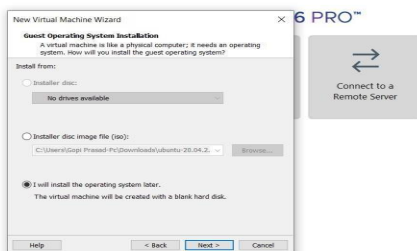
Step 3: Start the virtual machine.

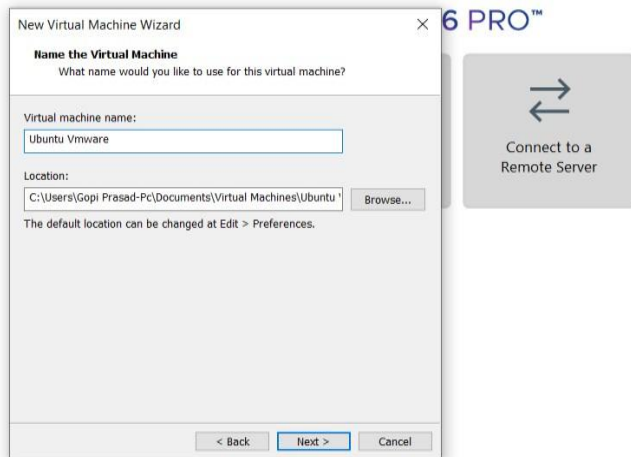
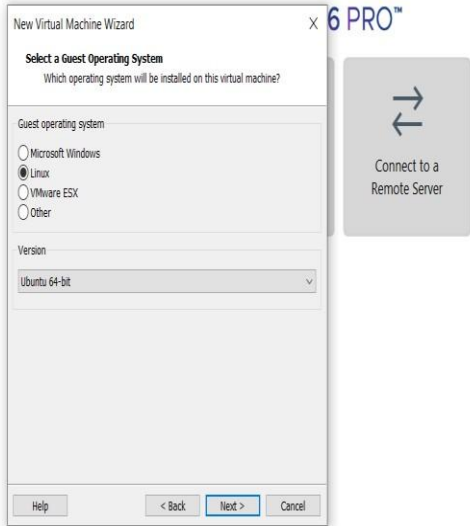


Step 4: Create or open another virtual machine with a different configuration.

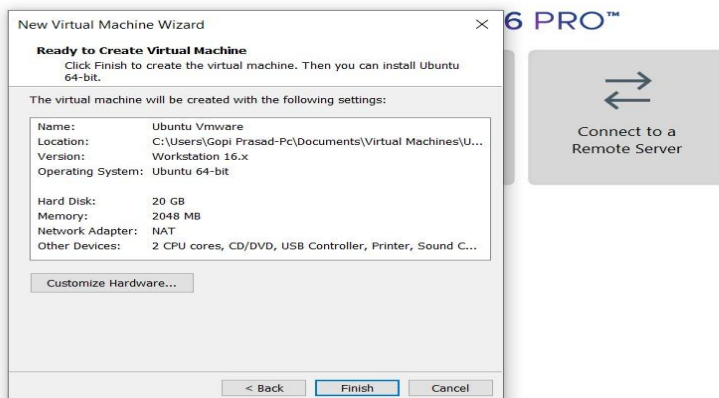


! Your evaluation period ends in 12 days. 1. Get a license key 2. Enter a license key

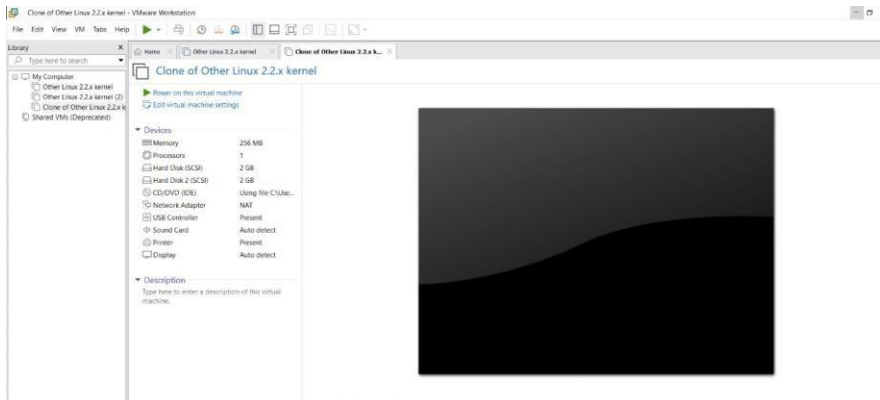




Step 5: Run multiple virtual machines simultaneously and observe the system resources.



Step 6: Check how many virtual machines can be utilized at the same time.



RESULT

Thus, virtual machines with different configurations were successfully run, and the number of virtual machines that could be utilized simultaneously was checked.